

ChessConnect – Classic Chess, Modern Connectivity

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Abstract

ChessConnect is a new and exciting chess experience that combines the iconic gameplay of traditional chess with online connectivity in a standalone package. ChessConnect allows users to play over-the-board games against an online opponent by interfacing with the chess website Lichess.org. Using magnetic pieces and an internal cartesian gantry, the online opponents' moves are automatically relayed onto the chess board in real time while preserving the traditional look and feel that serious players expect. Additionally, the ChessConnect user's moves are automatically detected and sent to the online game using magnetic detection switches, allowing for easy, seamless connectivity between the online platform and the user. All of this is done in a single, minimalistic device which, once linked to a Lichess.org account and connected to WiFi, requires no external devices and operates with only two buttons and an LCD screen. Using this technology, ChessConnect bridges the gap between traditional skill and modern convenience while removing the limitations of both formats.

Problem

Chess has evolved with the rise of online play, but this shift introduces compromises. Online distractions, shorter time controls, and the lack of physical interaction impact the development of good habits for over-the-board chess. However, finding local opponents of similar skill can be inconvenient, making online play the more accessible option despite its drawbacks. Many players struggle with translating online skills to physical games due to differences in coordination and board visualization.

Solution

ChessConnect bridges the gap between online and traditional formats, giving users the convenience of online play while maintaining the benefits of traditional chess. By integrating seamlessly with Lichess.org, users will be able to start a game against anybody in the world while playing the game on a physical board. Whenever a user makes a move on the board, it will automatically be interpreted and sent to the online game. Similarly, when the online opponent makes a move online, their piece will be automatically moved to its new position on the board using the internal gantry. By eliminating manual move translation and allowing for the convenience of online connectivity, ChessConnect offers a seamless and immersive chess experience.

ChessConnect (Internal & External View)



Important Components

Arduino Uno:

The Arduino Uno R4 WiFi microcontroller is the “brains” of ChessConnect. It is used to acquire data from the various inputs, interpret the user's chess moves, and connect to and facilitate online gameplay.

LCD Display:

The LCD will display the UI needed to setup a game and transition to show the in-game chess clock once the game has started. If an error in piece positions occurs, the LCD screen will also be used to inform the user how to resolve the error.

Reed Switches:

Reed switches are used to track the positions of the pieces during a game. These switches are located under every square of the board which allows them to interact with the magnets in the base of each piece.

Electromagnet:

The electromagnet is used in the XY rail system to control piece movement. When the online opponent's move is received, the magnet will be positioned beneath the piece, turn on, carry the piece to its new position, and turn off.

Motors:

Two motors are used to facilitate the movement of the electromagnet mounted on the XY gantry. This allows the ChessConnect to maneuver the game pieces from within the board.

Hardware Block Diagram

